

INDUSTRIAL ASSET GRAPH

Machine Documentation and Knowledge Preservation Project

First Machine Review Package

Repository	https://github.com/tobystrings/industrial-asset-graph
First asset	FG-L4-MTN-001 - META 150 HS case-forming machine
Document status	Working review draft - built to be corrected and expanded
Verification rule	Do not treat assumptions as confirmed facts
Primary reviewer	Experienced facility electrician
Revision	0.2 - July 30, 2026

Purpose: preserve the practical knowledge needed to operate, troubleshoot, repair, modify, and expand facility equipment.

1. Project Summary

We are building one central equipment-knowledge system for the facility. It will hold machine records, electrical information, controls, utilities, photographs, drawings, procedures, repair history, and the practical knowledge learned by experienced maintenance personnel.

The first machine is being documented completely so it can become the repeatable pattern for every other machine. The system must remain editable because field corrections, machine changes, replacements, and additions are inevitable.

What this project is meant to prevent

- Losing years of practical knowledge when an experienced electrician retires.
- Wasting time tracing the same wires, identifying the same parts, and rediscovering the same faults.
- Treating guesses, old drawings, or undocumented changes as verified facts.
- Keeping important information scattered across binders, phones, computers, and individual memory.

What the finished system should answer

- What panel and circuit feed this machine?
- What shuts this motor down?
- Which PLC input receives this sensor?
- What devices are in the safety circuit?
- Where are the disconnect and isolation points?
- What usually causes this alarm or failure?
- What part was used previously, and what was changed?
- Which information is confirmed and which still needs field verification?

2. Instructions for the Electrician Reviewing This

Please review this as a practical maintenance record, not as an engineering report. Mark up anything that is wrong, incomplete, or misleading.

- ☐ Correct machine names, component names, and functions.
- ☐ Cross out anything that is only a guess or is not true.
- ☐ Add the first things you normally check when the machine fails.
- ☐ Add recurring problems, odd behavior, and known weak points.
- ☐ Add startup, shutdown, reset, and recovery steps that are not in the manual.
- ☐ Identify important parts that should be kept on the shelf.
- ☐ Point out any modifications that do not match the original drawings.
- ☐ Mark anything that still needs to be traced, tested, photographed, or confirmed.

Most valuable information: the practical details that an experienced electrician knows but the OEM manual does not explain.

3. Evidence and Verification Rules

Every important fact should show how certain it is. Unknown information stays unknown until somebody verifies it.

State	Meaning	Use
Verified	Confirmed from reliable evidence.	Use as established machine information.
Field Verify	Expected to be correct but still needs inspection or testing.	Keep on the field checklist.
Inferred	Reasonable conclusion from photos or system layout.	Do not present as confirmed.
Disputed	Sources or people disagree.	Keep both positions until resolved.
Retired	No longer active or installed.	Keep for history and old-document reference.

Acceptable evidence

- Equipment nameplate or component label.
- Field photograph.
- Electrical or pneumatic drawing.
- Manufacturer manual.
- PLC program or HMI project.
- VFD or servo parameter file.
- Maintenance or CMMS record.
- Interview with experienced personnel.
- Direct testing, wire tracing, and panel inspection.

4. Current Progress - First Machine

Item	Current record
Asset ID	FG-L4-MTN-001 - shown on facility asset label
Machine	META 150 HS case-forming machine
Manufacturer	Smurfit-Stone Packaging Systems - based on visible machine identification
Serial number	MT081619A - requires final nameplate confirmation
Location	Line 4
Main supply	480 VAC, 3-phase, 60 Hz
Full-load current	34.2 A
Recommended branch protection	40 A
Control voltage	24 VDC
Compressed air	80 PSI, 30 SCFM
Observed controls	Allen-Bradley PowerFlex drive; Control Techniques servo drives; Bosch Rexroth inline I/O and power supply
Documentation collected	Machine photos, nameplate photos, HMI screens, and electrical-panel photos

Important caution

The exact function of several components and servo axes has not yet been confirmed. Descriptions based only on panel photographs must remain marked Field Verify or Inferred until drawings, labels, programs, or field operation confirm them.

5. Current Machine Summary

Purpose

The machine appears to form corrugated cartons from flat blanks. It uses powered motion, servo-controlled assemblies, pneumatics, vacuum, and adhesive-related functions. The exact production sequence must be reviewed with the electrician and operators.

Working operating sequence - Field Verify

- Flat corrugated blanks are loaded into the magazine.
- A blank is picked or pulled into the forming area.
- The blank is opened and positioned around the forming section or mandrel.
- Flaps are folded in the programmed sequence.
- Adhesive is applied where required.
- The joint or flaps are pressed long enough to hold.
- The finished carton is discharged to the next section.

Observed major control equipment

- One Allen-Bradley PowerFlex 70 variable-frequency drive, labeled VFD1.
- Three Control Techniques servo drives. Exact model numbers and axis assignments require close-up label photographs and field confirmation.

- Bosch Rexroth inline remote I/O modules.
- Bosch Rexroth 24 VDC power supply.
- Branch protective devices, contactors, control relays, terminal blocks, grounding, wire duct, and cabinet cooling.

6. Information Still Required

Electrical and controls

- ☐ Incoming disconnect manufacturer, model, rating, and upstream source.
- ☐ Exact panel, MCC, transformer, breaker, and circuit feeding the machine.
- ☐ PLC or controller manufacturer, model, firmware, and program-backup location.
- ☐ HMI manufacturer, model, software version, password-control method, and backup location.
- ☐ Complete remote-I/O module list, node information, and I/O map.
- ☐ VFD catalog number, motor served, horsepower, wiring, and parameter backup.
- ☐ Each servo-drive model, motor served, encoder type, home method, and parameter backup.
- ☐ Safety relay or safety-controller information and a verified safety-circuit description.
- ☐ Network devices, addresses, communication protocols, and cable routing.
- ☐ Terminal-strip map and field-wire destinations.
- ☐ Any undocumented jumpers, bypasses, field modifications, or abandoned wiring.

Motors, sensors, pneumatics, and mechanical items

- ☐ Every motor and gearbox nameplate, location, function, rotation, and driven assembly.
- ☐ Every photoeye, proximity switch, limit switch, pressure switch, vacuum switch, and safety switch.
- ☐ Every solenoid valve, cylinder, regulator, vacuum device, and normal pressure setting.
- ☐ Air-isolation point and method for releasing stored pneumatic energy.
- ☐ Belts, chains, bearings, lubrication points, tension guidance, and common wear items.
- ☐ Glue-system manufacturer, model, operating settings, isolation points, and common faults.

Operation and maintenance knowledge

- ☐ Normal startup and shutdown procedure.
- ☐ Restart after power loss or emergency stop.
- ☐ Recipe list and product/changeover information.
- ☐ Normal speed and important operator settings.
- ☐ Top recurring failures and quickest checks.
- ☐ Alarm list with plain-language causes and recovery steps.
- ☐ Preventive-maintenance schedule based on actual plant conditions.
- ☐ Critical spare parts and known substitute parts.
- ☐ Repair and modification history.
- ☐ Anything the retiring electrician considers important but undocumented.

7. Industrial Asset Graph Repository Integration

This machine package will become part of the existing Industrial Asset Graph repository. The repository is the facility's long-term industrial knowledge base, and the graph is one way to view equipment and its relationships.

Basic asset hierarchy

Facility -> Building -> Area -> Production Line -> Machine -> Assembly -> Panel or Component

Assets must also support relationships outside that hierarchy, such as: panel feeds VFD, VFD controls motor, sensor reports to PLC input, safety relay removes enable power, compressor supplies machine air, and upstream equipment enables downstream equipment.

Required machine package

- A human-readable document for electricians, mechanics, supervisors, contractors, training, and printed use.
- A machine-readable record for search, filtering, dependency tracing, reports, AI assistance, and future CMMS integration.
- Evidence files such as photographs, manuals, drawings, PLC backups, and VFD parameters.
- Relationships connecting the machine to panels, utilities, controls, components, and other equipment.
- A verification checklist separating confirmed information from assumptions.

Proposed machine folder pattern

```
machines/<machine-id>/
  README.md
  asset.json
  relationships.json
  electrical.md
  controls.md
  operation.md
  troubleshooting.md
  loto.md
  parts.md
  verification-checklist.md
  photos/
  drawings/
  manuals/
  plc/
  vfd/
```

Development rules

- Inspect the current repository before changing its architecture.
- Preserve working functions and map new data into existing schemas where possible.
- Keep machine data separate from application code.
- Extend schemas only when needed and validate required fields and broken relationships.
- Keep fictional or sample data clearly separated from verified facility data.
- Favor plain labels and practical navigation over unnecessary software complexity.
- Do not build the first machine as a one-off static screen.

8. Change and Revision Rules

Every correction or alteration should be traceable. The system should preserve the current value and enough history to understand what changed and why.

- Date of change.
- Person making the change.
- Reason for change.
- Previous value.
- New value.
- Evidence supporting the change.
- Review and approval status.

Date	Changed by	Item	Old value	New value	Evidence	Approved

9. Immediate Next Steps

1. Review this package with the experienced electrician.
2. Correct inaccurate information and mark every uncertain statement.
3. Walk the machine and photograph all missing labels, motors, sensors, valves, and isolation points.
4. Locate the electrical drawings, manuals, PLC/HMI files, and drive parameters.
5. Record the electrician's common-failure and troubleshooting knowledge.
6. Create the approved human-readable machine record.
7. Convert approved facts into structured asset and relationship records.
8. Validate the machine package and add it to the Industrial Asset Graph.
9. Use the completed package as the template for the next machine.

10. Electrician Field Review Notes

Machine: FG-L4-MTN-001 - META 150 HS

What is wrong or misleading?

Common failures and first checks

Startup, shutdown, and reset knowledge

Known modifications or drawing errors

Parts we should keep in stock

Other information that must not be lost

Review Signoff

Reviewed by		Date	
Corrections entered by		Date	
Approved for repository		Revision	